PoI. After all, points that are around the current incumbent may improve with high probability, but their improvements would likely be minimal (which is what we empirically observed in our code).

A point that is far away, which we don't know much about, might give a lower improvement probability, but because there's a chance that this point will give a large improvement, EI might assign a higher score to it. In other words, while PoI might be considered a *risk-averse* BayesOpt policy that cares about improving from the incumbent, however small the improvement is, EI balances between the risk and reward in order to find the point that best balances the tradeoff. This is illustrated in Figure 4.16, which compares PoI and EI using the same data set and trained GP.

We see that the candidate data point chosen by PoI (around 0.6) is different from one chosen by EI (around -0.7). The former is close to the current incumbent, so it's likely that querying it will help us improve. However, EI sees that there's more uncertainty in other regions far away from the incumbent, which may lead to a larger improvement. Thanks to this reasoning, EI favors data points that better balance between exploration and exploitation.

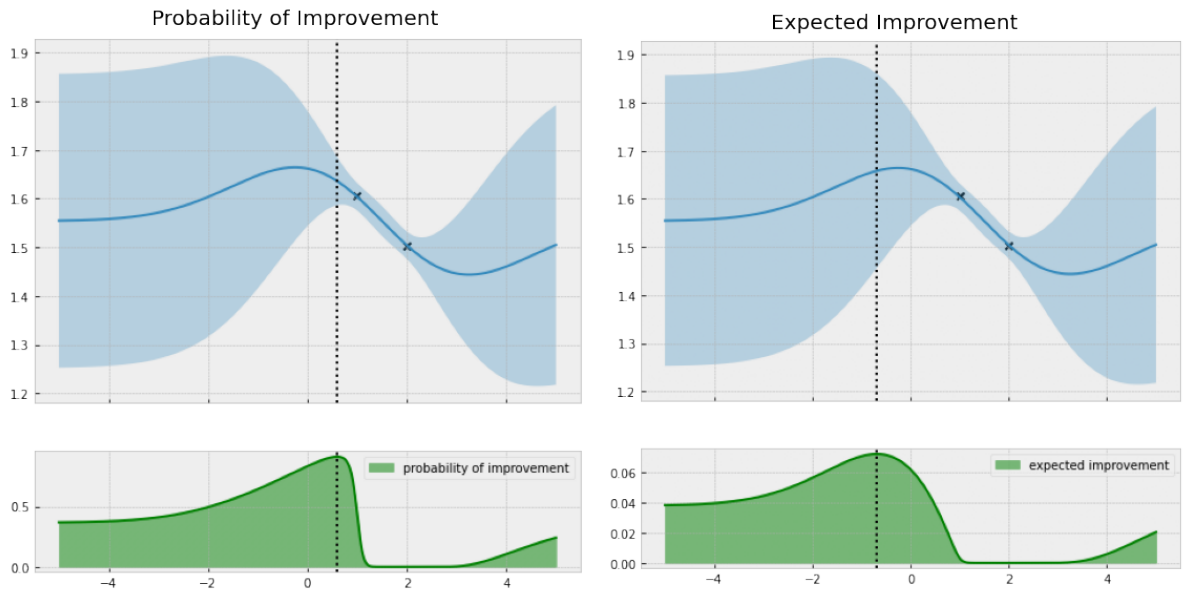


Figure 4.16 The difference between Probability of Improvement (left) and Expected Improvement (right). Expected Improvement balances exploration and exploitation better.

Another nice property of EI that showcases this balance is in the way it assigns the acquisition scores to data points with the same predictive mean or standard deviation. Specifically:

- If two data points have the same predictive mean but different predictive standard deviations, then the one with the higher uncertainty will have a higher score. The policy therefore rewards exploration.
- If two data points have the same predictive standard deviations but different predictive means, then the one with the higher mean with have a higher score. The policy therefore

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>